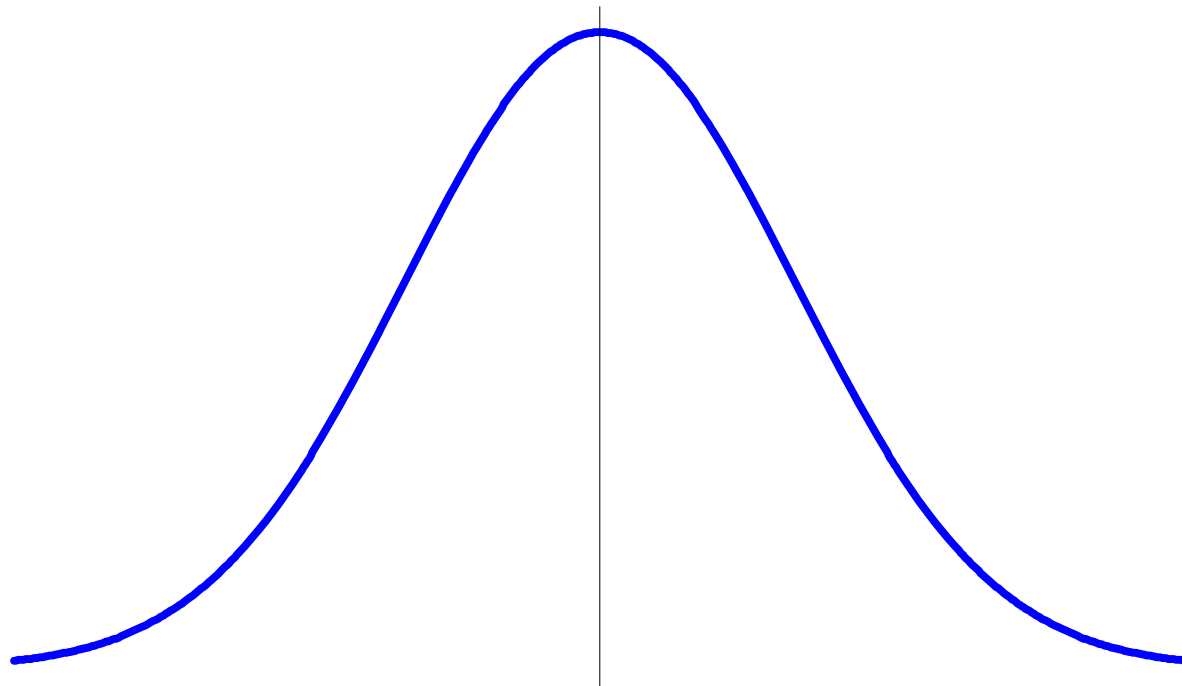


Conditions for Hypothesis tests

- **Independence:** Sampled observations must be independent (random sample)
- if sampling without replacement, sample size should be $n < 10\%$ of population
- **Sample size/skew:** sample size should be larger than is $n \geq 30$, if the population distribution is very skewed (more data are taking on lower or higher values).

Normal or z-test: *The distribution of data is assumed to be normal distribution of symmetric curve*

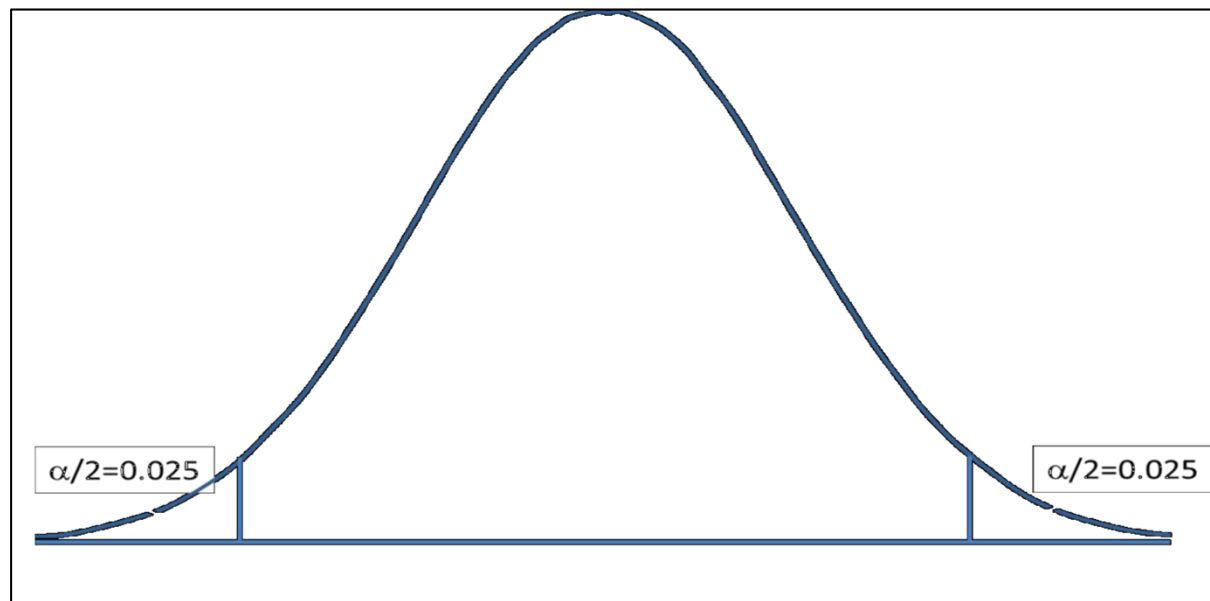


Normal or z-test: Conditions

- z-test is preferable when sample size, n is greater than 30 that is $n > 30$
- population variance (denoted by σ^2) is generally known before
- the variances of the different samples should be the same
- all individuals must be selected at random from the population
- For two samples test, sample sizes should be as equal as possible but some differences are allowed

Two tailed test for z-test ($H_A : \mu \neq \mu_0$)

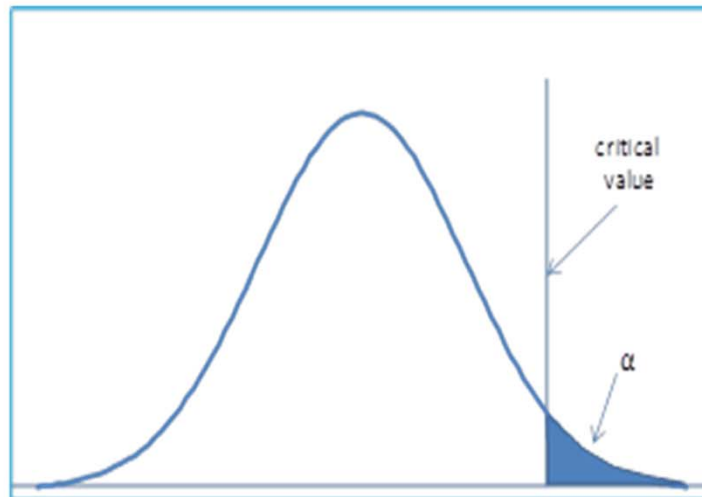
- In a two-tailed z-test, the level of significance, α is split across the two tails
- Let us assume that the level of significance, $\alpha = 0.05$, after splitting into two parts, each tail has a the level of significance of 0.025.
- We will reject the null hypothesis if your sample mean falls in either tail of the distribution.



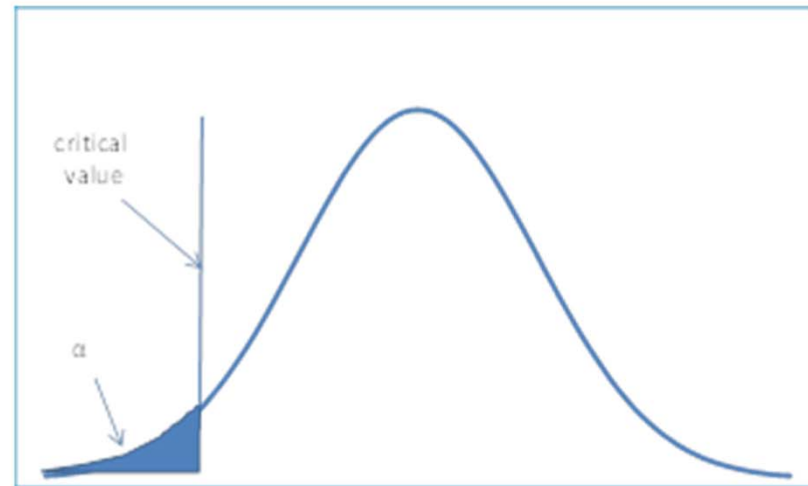
One tailed test for z-test (right-tailed $H_A : \mu > \mu_0$ or left-tailed $H_A : \mu < \mu_0$)

We would use a single-tail hypothesis test when the direction (greater or less than) of the results is anticipated or we are only interested in one direction of the results. In a one-tailed z-test, the alpha level is not split across the two tails. A single-tail hypothesis test also means that we have only one critical region because we put the entire critical region into just one side of the distribution. When the alternative hypothesis is that the sample mean is greater, the critical region is on the right side of the distribution (see below). When the alternative hypothesis is that the sample is smaller, the critical region is on the left side of the distribution.

Right-tailed

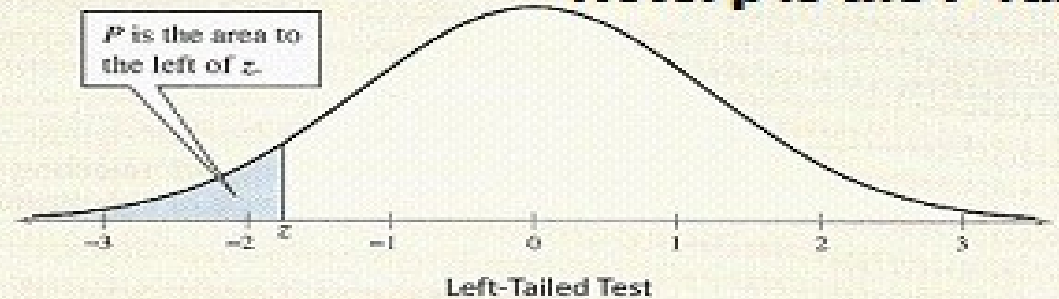


Left-tailed

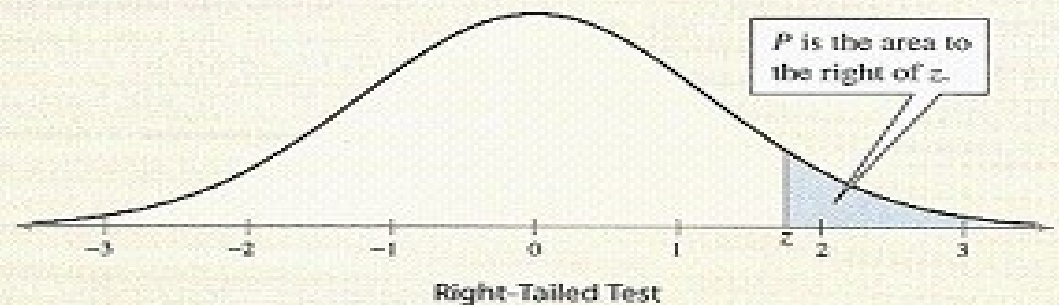


1. If the alternative hypothesis H_a contains the less-than inequality symbol ($<$), the hypothesis test is a **left-tailed test**.

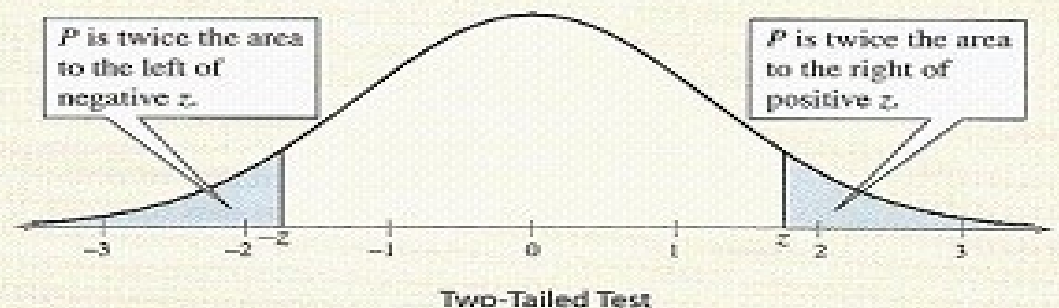
Note: p is the P-value



2. If the alternative hypothesis H_a contains the greater-than inequality symbol ($>$), the hypothesis test is a **right-tailed test**.



3. If the alternative hypothesis H_a contains the not-equal-to symbol ($\neq$), the hypothesis test is a **two-tailed test**. In a two-tailed test, each tail has an area of $\frac{1}{2}P$.



Z-Test's for Different Purposes(uses)

- 1. *Independent one-Sample mean-Test:*** An independent one-sample Z-test is used to test whether the average of a sample differ significantly from a population mean.
- 2. *Independent Two-Sample mean-Test:*** The independent two-sample z-test is used to test whether two population means are significantly different from each other, using the means from randomly drawn samples
- 3. *single proportion:*** z test is used to test a hypothesis on a specific single value of the population proportion.(say 50% people like tea)

Z-Test's for Different Purposes

4. Difference of proportions: z test is used to test the hypothesis whether the two populations have the same proportion. For example suppose one is interested to test if there is any significant difference in proportion (or %) for the habit of tea drinking between male and female of a town. In such a situation, z-test for difference of proportions can be applied. One would have to obtain two independent samples from the town- one from males and the other from females and determine the proportion of tea drinkers in each sample in order to perform this test.

5. Dependent z-Test for Paired Samples: The dependent z-test for paired samples is used when the samples are paired. One sample has been tested twice (repeated measures), e.g, weight of persons before and after exercise or diet or taking medicine

- There are two samples that are the same size.
- The two samples are **dependent**. (This can be when the two similar subjects are matched or paired, or when subjects come from the same source, or when 2 observations are made on the same subject such as pre and post tests.)

One Sample z-Test for the Population Mean

When can this test be used?

- The population is normally distributed or the sample size is large enough that the mean is normally distributed.
(A rule of thumb is that the sample size is large enough if $n \geq 30$.)
- The population standard deviation, σ is **known**.

Statistical Hypothesis Testing

Normal or z-test

Steps to do this test:

1. State the hypotheses:

$$H_0: \mu = \mu_0$$

$$H_A: \mu \neq \mu_0 \quad (\text{or } H_A: \mu > \mu_0 \text{ or } H_A: \mu < \mu_0)$$

2. Pick a significance level, α .

3. Compute the test statistic:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

4 Calculate critical region(Critical region approach):

find the tabulated value of z and then find critical region as

For two tailed critical region : $C.R \geq Z(\text{tabulated})$ or $C.R \leq Z(\text{tabulated})$

For right- tailed ($H_A: \mu > \mu_0$) critical region : $C.R \geq Z(\text{tabulated})$

For left-tailed ($H_A: \mu < \mu_0$) critical region : $C.R \leq Z(\text{tabulated})$

Also we can use p-value

5.State the conclusion:

if the calculated value of z falls in the critical region (or $p\text{-value} < \alpha$) we reject null hypothesis, we accept the alternative.

Steps to do this test:

p value approach

or 4. Find the p-value: Find the probability that Z is beyond (more extreme than) your test statistic

The p-value depends on which alternative hypothesis is being used.

Where z is the calculated value from test statistic and Z we have search in table

If $H_A: \mu \neq \mu_0$, p-value = $2P(Z \geq |z|)$ [in R, $> 2*(1-pnorm(|z|))$

[XL=2*(1-NORM.S.DIST(|z|, TRUE))]

If $H_A: \mu > \mu_0$, p-value = $P(Z \geq z)$ (in R, $> (1-pnorm(z))$)/[XL=1-NORM.S.DIST(z, TRUE)]

If $H_A: \mu < \mu_0$, p-value = $P(Z \leq z)$ (in R, $> pnorm(z)$) .[XL=NORM.S.DIST(z, TRUE)]

It is noted that most of the software provided p value

5.State the conclusion:

For small value, that is p-value $< \alpha$, which means it is unlikely or very less chance to obtain better result than the observed result of test statistics for the given sample and thus result goes in favour of H_A

Therefore, the p-value $< \alpha$, **we** reject the null hypothesis H_0

and accept the alternative hypothesis.

Statistical Hypothesis Testing

Normal or z-test

Remember-Tabulated value of Z

Level of significance α	Two-tailed Z	Right-tailed Z	Left-tailed Z
0.10 (10%)	± 1.645	1.282	-1.282
0.05 (5%)	± 1.960	1.645	-1.645
0.010 (1%)	± 2.576	2.326	-2.326

Example 1:(One Sample z-Test for the Population Mean)

(Non directional tests, or **two-tailed tests**, are hypothesis tests where the alternative hypothesis is stated as *not equal to* ($\neq$): It is reported that the population mean score on the quantitative portion of the Graduate Record Examination (GRE) General Test for students taking the exam between 1994 and 1997 was with mean 558 and standard deviation 139 . Suppose we select a sample of 100 participants ($n = 100$). We record a sample mean equal to 585. Compute the one-independent sample z test for whether or not we will retain the null hypothesis (mean = 558) at 5% level of significance ($\alpha = .05$).

Statistical Hypothesis Testing

Normal or z-test

Step 1: State the hypotheses.

$H_0: \mu = 558$ (Mean test score is 558 in the population)

$H_A: \mu \neq 558$ (Mean test score is not 558 in the population)

Step 2: Specification of the level of significance, $\alpha = 0.05$.

Step 3: Compute the test statistic z as

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Here sample mean, $\bar{x} = 585$, population mean, $\mu = 558$, s.d = 139, $n = 100$

$Z = (585 - 558) / (139 / \sqrt{100}) = 27 / 13.9 = 1.94$

Step 4: Calculation tabulated value and critical region:

this is two-tailed test the tabulated value of z(tab) at 5% level of significance is 1.96

$C.R \geq 1.96$

Step 5: Statistical decision:

Since the calculated value of $|z| = 1.94 < 1.96$, does not fall in the critical region, we fail to reject the null hypothesis at **5% level of significance that**

We conclude that the mean score on the GRE General Test in this population is 558 (the value stated in the null hypothesis).

P-value approach

Calculation of the p value: The probability, p-value for $H_A: \mu \neq 558$, from z table, the value is .0262 [for two tailed $\text{prob}(z > 1.94)$ or $\text{prob}(z < -1.94)$ or

simple $p(z > 1.94) = 1 - p(z < 1.94) = 1 - 0.9738 = 0.0262$.[in excel, =1-NORM.S.DIST(1.94,TRUE)]

Since this is a two tailed test, we multiply .0262 times 2: $p = (.0262) \times 2 \text{ tails} = .0524$.

As we got the p value of 0.0524, which is larger than 0.05, then

- It is "likely" (as p value, $0.0524 > 0.05$) that we would observe an extreme value or better result than the test statistics $z = 1.94$ for the given sample and thus result of test statistics does not go in favour of H_A

- In other words, the calculated test value $z = 1.94$ is not satisfactory, still we have chance to get better than the observed difference of test value for the given sample

* Any statistical package (say, SPSS,R, Python) the p-value is already calculated as 0.0524

Step 5: Statistical decision:

we know that when p-value: $p < 0.05$, we reject null hypothesis

Since p-value, $0.0524 > 0.05$, we do not reject the null hypothesis that is we accept it

We conclude that the mean score on the GRE General Test in this population is 558 that is the score founded from the sample (585) is not significantly differ from the claimed score(the value stated in the null hypothesis).

Example 2: A tire manufacturing plant produces 15.2 tires per hour. This yield has an established variance of 2.5. New machines are recommended, but will be expensive to install. Before deciding to implement the change, 32 new machines are tested. They produce 16.1 tire per hour. Is it worth buying the new machines?

1. Formulate hypotheses:

$$H_0 : \mu = 15.2$$

(ie. Mean yield of new machines is equal to 15.2 with a variance of 2.5)

$$H_1 : \mu > 15.2$$

(ie. Mean yield of new machines is greater than 15.2)

2.Choose α : We choose $\alpha = 0.010$.

3. Select and Compute the test statistic: Here we must use the z statistic to test the null hypothesis since the variance is known

Then the value of z

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Here sample mean, $\bar{x} = 16.1$, population mean, $\mu=15.2$, s.d $=\sigma =\text{sqrt}(2.5)=1.58$, $n=32$

¹⁶ $z = 3.22$

4. Find the critical region: The z-value obtained from z-Table 1 for α at 1% level of significance 2.33. Hence, the critical region for a one tailed (right-tailed) test is:

$$C.R > 2.33$$

5. Conclusion: Since the calculated of $z = 3.22 > 2.33$, which falls in the critical region, we reject the null hypothesis and accept the alternative. It is highly statistically significant that the mean yield from the new machines is greater than 15.2 at 1% level of significance. Thus the mean yield of the new machine is greater than 15.2

can you try for p-value approach

p-value

$$p(z > 3.22) = 1 - p(z < 3.22) = 1 - 0.9994 = 0.0006$$

$$p\text{-value}, 0.0006 < 0.01,$$

It is “unlikely” that we would observe an extreme value or better result than the test statistics $z=3.22$ for the given sample and thus result of test statistics goes in favour of H_1 . We reject the null hypothesis and accept the alternative. It is highly statistically significant that the mean yield from the new machines is greater than 15.2 at 1% level of significance.

Examples of z-test for one sample mean

1. A rental car company claims the mean time to rent a car on their website is 60 seconds with a standard deviation of 22 seconds. A random sample 36 customers attempted to rent a car on the website. The mean time to rent was 55 seconds. Is this enough evidence that the mean time is less than the company's claim? Consider $\alpha=0.05$. [calculated, $z=-1.36$, $C.R < -1.645$, $p\text{-value}=0.0853$]
2. A teacher wanted to know how well the gifted students in here class perform relative to her other classes. She administers a standardized test with a mean of 50 and standard deviation of 10. Her class of 31 students has an average score of 55. Test that their average score higher than 50?
3. A researcher wanted to study the effects of mentoring on intelligence scores. He wanted to know as a baseline what the average intelligence of his students were relative to the general population. He used a standardized IQ test which has a mean of 100 and standard deviation of 16. The 50 students in his study scored an average of 102 on the IQ test. Does his study support the baseline?